

IMPLEMENTATION OF ALGORITHMS FOR BELIEF CHANGE IN BELIEF BASES

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Abstract

Four types of contraction operators on belief bases are presented and implemented. Partial meet contraction selects, among the maximal subsets of the base that fail to imply the sentence being contracted, those chosen by a selection function, and intersects them. Kernel contraction instead identifies the minimal subsets (kernels) that imply the sentence, and removes from the base at least one element of each kernel, as determined by an incision function. Smooth kernel contraction is a more conservative refinement of kernel contraction, obtained by requiring the incision function to satisfy an additional smoothness condition that preserves the inferential content of the base. Finally, basic AGM-generated base contraction obtains its outcome by applying a basic AGM contraction operator to the logical closure of the base and intersecting the result with the original base. Each operator was implemented in Python, using the Z3 theorem prover to verify logical deduction by refutation and the `itertools` library to generate the subsets required by the underlying algorithms, together with a dedicated parser for propositional logic formulas and a command-line interface for entering belief bases and formulas and displaying the corresponding contraction results.

Keywords:

Kernel contraction, Partial Meet Contraction, Basic AGM generated base contraction, Computational implementation.

Chapter 1

Introduction

Belief Change is a multidisciplinary field concerned with how a rational agent modifies its beliefs when new information becomes available. The most influential framework in this area is the AGM model [AGM85], in which beliefs are represented as logical sentences and an agent's epistemic state is modelled by a belief set, that is, a logically closed set of sentences. Among the various belief change operations, contraction has received considerable attention. Informally, a contraction operation removes beliefs from a belief set so that a specified sentence is no longer derivable from it. In the AGM framework, contraction operators are characterized by a collection of rationality postulates that define their expected behaviour. The following postulates, introduced in [AGM85], characterize contraction operators on a belief set K.

$(K-1)$ $K - \alpha = Cn(K - \alpha)$.	(Closure)
$(K-2)$ $K - \alpha \subseteq K$.	(Inclusion)
$(K-3)$ If $\alpha \notin K$, then $K - \alpha = K$.	(Vacuity)
$(K-4)$ If $\alpha \notin Cn(\varnothing)$, then $\alpha \notin K - \alpha$.	(Success)
$(K-5)$ If $\alpha \leftrightarrow \beta \in Cn(\varnothing)$, then $K - \alpha = K - \beta$.	(Extensionality)
$(K-6)$ $K \subseteq Cn((K - \alpha) \cup \{\alpha\})$.	(Recovery)
$(K-7)$ $(K - \alpha) \cap (K - \beta) \subseteq K - (\alpha \wedge \beta)$.	(Conjunctive overlap)
$(K-8)$ $K - (\alpha \wedge \beta) \subseteq K - \alpha$ whenever $\alpha \notin K - (\alpha \wedge \beta)$.	(Conjunctive inclusion)

These postulates are usually designated by *AGM postulates for contraction*. Postulates $(K-1)$—$(K-6)$ are called *basic AGM postulates for contraction* and an operator $-$ that satisfies those properties is called a *basic AGM contraction*. Postulates $(K-7)$ and $(K-8)$ are designated by *supplementary AGM postulates for contraction*.

Although the AGM model has become the standard approach to belief revision, several authors (see [FH11] for a survey) have identified limitations that reduce its applicability in certain settings. In response, a variety of extensions and generalizations of the AGM framework have been introduced. One of the main criticisms concerns the representation of an agent's epistemic state as a logically closed set of

beliefs, commonly referred to as a belief set. This representation presents a number of drawbacks. Since belief sets are typically very large, and may even be infinite, they are impractical for computational implementations. Beyond these computational issues, the requirement of logical closure has also been criticized on conceptual grounds. As noted by Rott [Rot00], the AGM framework assumes epistemic agents to be "ideally competent regarding matters of logic," meaning that they accept every logical consequence of their beliefs and always maintain a consistent belief set. Such assumptions are generally regarded as unrealistic, particularly when modeling human reasoning. In addition, representing beliefs as logically closed sets obscures important distinctions: it does not differentiate between different inconsistent belief states, nor does it distinguish basic (explicit) beliefs from those that are merely inferred from them. Several existing contraction operators for belief sets have been adapted to scenarios where the belief state of an agent is represented by a belief base – a set of sentences that is not necessarily logically closed. For example, the partial meet contractions for belief bases were presented in [Han91, Han92, Han93] and the kernel contractions were introduced in [Han94].

1.1 Report Structure

The remainder of this document is organized as follows. Chapter 2 recalls the theoretical background on belief base contraction: it presents the main rationality postulates for base contraction and reviews four constructive models of contraction operators (partial meet contraction, kernel contraction, smooth kernel contraction, and basic AGM-generated base contraction) together with their axiomatic characterizations. Chapter 3 describes the computational implementation of these operators: it details the development environment and tools used, the verification of logical consequence by refutation, the generation of remainder sets, and the algorithms implemented for each of the contraction operators presented in Chapter 2.

1.2 Formal Preliminaries

We will assume a propositional language $\mathcal{L}$ that contains the usual truth functional connectives: $\neg$ (negation), $\wedge$ (conjunction), $\vee$ (disjunction), $\rightarrow$ (implication) and $\leftrightarrow$ (equivalence). We shall make use of a consequence operation Cn that takes sets of sentences to sets of sentences and which satisfies the standard Tarskian properties, namely *inclusion, monotony* and *iteration*. Furthermore we will assume that Cn satisfies *supraclassicality, compactness* and *deduction*. We will call $Cn(A)$ the *logical closure* of A. We will sometimes use $Cn(\alpha)$ for $Cn(\{\alpha\})$, $A \vdash \alpha$ for $\alpha \in Cn(A)$, $\vdash \alpha$ for $\alpha \in Cn(\varnothing)$, $A \nvdash \alpha$ for $\alpha \notin Cn(A)$, $\nvdash \alpha$ for $\alpha \notin Cn(\varnothing)$. The letters $\alpha, \beta, \ldots$ (except for γ and σ) will be used to denote sentences of $\mathcal{L}$. $A, B, \ldots$ shall denote sets of sentences of $\mathcal{L}$. K is reserved to represent a set of sentences that is closed under logical consequence (i.e. $K = Cn(K)$) — such a set is called a *belief set* or *theory*. Given $A \subseteq \mathcal{L}$, we shall say that $-$ is a (contraction) operator (or function) on A when $-$ is a function such that $- : \mathcal{L} \longrightarrow P(\mathcal{L})$. Furthermore, in that context,

we shall represent by $A - \alpha$ the image of a sentence α by $-$. Given a set A we will denote the power set of A by $\mathcal{P}(A)$.

Chapter 2

Contraction Operators on Belief Bases

In this chapter we recall some well known postulates for base contraction and also several constructive models of contraction functions on belief bases.

2.1 Postulates for base contraction

The following are some well known postulates for belief base contraction:

Success: If $\not\vdash \alpha$, then $A - \alpha \not\vdash \alpha$.
Inclusion: $A - \alpha \subseteq A$.
Failure: If $\vdash \alpha$ then $A \div \alpha = A$.
Vacuity: If $A \not\vdash \alpha$, then $A \subseteq A \div \alpha$.
Relative Closure: $A \cap Cn(A \div \alpha) \subseteq A \div \alpha$.
Relevance: If $\beta \in A$ and $\beta \notin A \div \alpha$ then there is some set A' such that $A \div \alpha \subseteq A' \subseteq A$ and $\alpha \notin Cn(A')$ but $\alpha \in Cn(A' \cup \{\beta\})$.
Core-retainment: If $\beta \in A$ and $\beta \notin A \div \alpha$ then there is some set A' such that $A' \subseteq A$ and $\alpha \notin Cn(A')$ but $\alpha \in Cn(A' \cup \{\beta\})$.
Disjunctive elimination: If $\beta \in A$ and $\beta \notin A \div \alpha$, then $A \div \alpha \not\vdash \alpha \vee \beta$.
Extensionality: If $\vdash \alpha \leftrightarrow \beta$, then $A \div \alpha = A \div \beta$.
Uniformity: If it holds for all subsets A' of A that $\alpha \in Cn(A')$ if and only if $\beta \in Cn(A')$, then $A \div \alpha = A \div \beta$.

2.2 Belief Base Contraction Operators

In this section we recall some explicit definitions of contraction operators as well as axiomatic characterizations for them.

2.2.1 Partial meet contractions

The first kind of contraction operators that we will present are known as *partial meet contractions* and were originally presented in [AGM85]. We start by recalling the concept of *remainder set*, that is a set of maximal subsets (of a given set) that fail to imply a given sentence. Formally:

Definition 1 ([AM81]) *Let A be a belief base and α a sentence. The set $A \perp \alpha$ (A remainder α) is the set of sets such that $B \in A \perp \alpha$ if and only if:*

1. $B \subseteq A$.
2. $B \not\vdash \alpha$.
3. *If $B \subset B' \subseteq A$, then $B' \vdash \alpha$.*

$A \perp \alpha$ is called remainder set of A by α and its elements are called remainders (of A by α).

The partial meet contractions are obtained by intersecting some elements of the (associated) remainder set. The choice of those elements is performed by a *selection function*.

Definition 2 ([AGM85]) *Let A be a set of sentences. A selection function for A is a function $\gamma : \{A \perp \epsilon : \epsilon \in \mathcal{L}\} \longrightarrow \mathcal{P}(\mathcal{P}(A))$ such that for all sentences α:*

1. *If $A \perp \alpha \neq \varnothing$, then $\gamma(A \perp \alpha)$ is a non-empty subset of $A \perp \alpha$;*
2. *If $A \perp \alpha = \varnothing$, then $\gamma(A \perp \alpha) = \{A\}$.*

Definition 3 ([AGM85, Han91]) *The partial meet contraction operator on A based on a selection function $\gamma : \{A \perp \epsilon : \epsilon \in \mathcal{L}\} \longrightarrow \mathcal{P}(\mathcal{P}(A))$ is the operator $\div_\gamma$ such that for all sentences α:*

$$A \div_\gamma \alpha = \cap\gamma(A \perp \alpha).$$

An operator $\div$ for a set A is a partial meet contraction if and only if there is a selection function γ for A such that $A \div \alpha = A \div_\gamma \alpha$ for all sentences α.

Hansson characterized partial meet contractions, defined on belief bases, in terms of postulates:

Observation 1 ([Han91]) *Let A be a belief base. An operator $\div$ on A is a partial meet contraction function for A if and only if $\div$ satisfies success, inclusion, uniformity and relevance.*

There are two limiting cases of partial meet contractions:

Definition 4 ([AGM85]) *Let $\div$ be an operator on a set A. Then:*

(a) $\div$ is an operator of maxichoice contraction if and only if it coincides with a partial meet contraction $\div_\gamma$ such that for all sentences α, $\gamma(A \perp \alpha)$ has exactly one element.

(b) $\div$ is an operator of full meet contraction if and only if it coincides with a partial meet contraction $\div_\gamma$ such that for all sentences α, if $A \perp \alpha$ is non-empty, then $\gamma(A \perp \alpha) = A \perp \alpha$.

2.2.2 Kernel contraction

As we mentioned previously the partial meet contraction operators of a given set by a sentence α are based on a selection among the maximal subsets of that set that fail to imply α. Another different proposal consists of constructing a contraction operator based on a selection of elements of A that are fundamental in some deduction of α and then discarding them when contracting A by α. Following this approach, Hansson, in [Han94], introduced the *kernel contraction* operators, which can be seen as a generalization of the *safe contraction* defined by Alchourrón and Makinson in [AM85].[1]

Definition 5 ([Han94]) *Let A be a set of sentences and α be a sentence. Then $A\perp\!\!\!\perp\alpha$ is the set such that $B \in A\perp\!\!\!\perp\alpha$ if and only if:*

1. $B \subseteq A$.
2. $B \vdash \alpha$.
3. *If* $B' \subset B$ *then* $B' \nvdash \alpha$.

$A\perp\!\!\!\perp\alpha$ is called the kernel set of A with respect to α and its elements are the α-kernels of A.

When contracting a belief α from a set A we must give up sentences of each α-kernel, otherwise α would continue being implied by the outcome of the contraction. The so-called incision functions [Han94] select the beliefs to be discarded.

Definition 6 ([Han94]) *Let A be a set of sentences. Let $A\perp\!\!\!\perp\alpha$ be the kernel set of A with respect to α. An incision function σ for A is a function such that for all sentences α:*

1. $\sigma(A\perp\!\!\!\perp\alpha) \subseteq \bigcup(A\perp\!\!\!\perp\alpha)$.
2. *If* $\varnothing \neq B \in A\perp\!\!\!\perp\alpha$, *then* $B \cap \sigma(A\perp\!\!\!\perp\alpha) \neq \varnothing$.

Definition 7 ([Han94]) *Let A be a set of sentences and σ an incision function for A. The kernel contraction on A based on σ is the operator $\div_\sigma$ such that for all sentences α:*

$$A \div_\sigma \alpha = A \setminus \sigma(A\perp\!\!\!\perp\alpha).$$

An operator $\div$ for a set A is a kernel contraction if and only if there is an incision function σ for A such that $A \div \alpha = A \div_\sigma \alpha$ for all sentences α.

Hansson presented in [Han94] an axiomatic characterization for kernel contractions defined on belief bases.

[1] For a deep study of safe contraction functions see [RH14].

Observation 2 ([Han94]) *Let A be a belief base. An operator $\div$ on A is a kernel contraction if and only if it satisfies success, inclusion, uniformity and core-retainment.*

Observation 3 *[BMVW12] Let A be a belief base. An operator $\div$ on A such that (i) if $A \perp \alpha = \varnothing$, then $A \div \alpha = A$ and $\bigcap(A \perp \alpha) \subseteq A \div \alpha \subseteq X$, for some $X \in A \perp \alpha$ is a kernel contraction.*

Sometimes, when contracting a set by means of a kernel contraction, some beliefs are removed without reason. For example if $\beta \in A$ and $\beta \in Cn(A \div \alpha)$, then β should also be in $A \div \alpha$. This can be solved if we ensure that an operator of kernel contraction satisfies *relative closure*. Hansson proposed in [Han94] a more conservative type of kernel contractions, that he designated by *smooth kernel contractions*. A *smooth kernel contraction* is a kernel contraction that is based on an incision function that satisfies the condition expressed in the following definition.

Definition 8 ([Han94]) *An incision function σ for a set A is smooth if and only if it holds for all subsets A' of A that if $A' \vdash \beta$ and $\beta \in \sigma(A \perp\!\!\!\perp \alpha)$ then $A' \cap \sigma(A \perp\!\!\!\perp \alpha) \neq \varnothing$. A kernel contraction is smooth if and only if it is based on a smooth incision function.*

The following observation presents an axiomatic characterization for smooth kernel contractions.

Observation 4 ([Han94]) *Let A be a belief base. An operator $\div$ on A is a smooth kernel contraction if and only if it satisfies success, inclusion, uniformity, core-retainment and relative closure.*

2.2.3 Basic AGM-generated base contraction

We now recall the definition of another kind of base contraction functions as well as an axiomatic characterization for them.

Definition 9 ([FKR08]) *Let A be a belief base. An operator $\div$ for A is a basic AGM-generated base contraction*[2] *if and only if, for all $\alpha \in \mathcal{L}$:*

$$A \div \alpha = (Cn(A) - \alpha) \cap A$$

where $-$ is a basic AGM contraction (i.e. an operator that satisfies the basic AGM postulates for contraction) on $Cn(A)$.

Observation 5 ([FKR08]) *Let A be a belief base. An operator $\div$ on A is a basic AGM-generated base contraction if and only if it satisfies success, inclusion, vacuity, extensionality and disjunctive elimination.*

[2] In [FKR08] these operators were designated by basic related-AGM base contractions.

Chapter 3

Implementation of Algorithms for Belief Base Contractions

3.1 Development environment and deployment

The system was developed using the Python programming language (version 3.10 or higher). To ensure the correct execution of the logical and structural operations, the following libraries and tools were utilized during development:

- **Z3 Theorem Prover (`z3-solver`)**:[1] An automated theorem prover developed by Microsoft Research. It serves as the core logical engine of the application, responsible for evaluating boolean satisfiability, manipulating symbolic variables, and verifying logical consequence by refutation.

- **`itertools` (Python Standard Library)**:[2] Used extensively to compute combinations and efficiently generate the powerset of the belief base.

- **`ast` and `re` (Python Standard Libraries)**:[3] Used to parse user input and translate natural logical syntax (e.g., $p \rightarrow q$) into internal Z3 representations, by traversing and modifying the Abstract Syntax Tree.

Executable Deployment:
To greatly improve user experience and accessibility, the final application was packaged using *PyInstaller*[4]. This tool compiles the modular Python scripts, the Z3 engine, and all necessary dependencies into a single, standalone executable file (`.exe`). Consequently, the end-user can run the application directly on their operating system without requiring a local Python environment or any library installations.

[1] `https://github.com/Z3Prover/z3`
[2] `https://docs.python.org/3/library/itertools.html`
[3] `https://docs.python.org/3/library/ast.html` and `https://docs.python.org/3/library/re.html`
[4] `https://pyinstaller.org/`

3.2 Logical consequence by refutation

The first important operation is the verification of logical consequence. In the implemented system, this is achieved by refutation: a set of premises B implies a formula α if and only if $B \cup \{\neg\alpha\}$ is unsatisfiable.

3.3 Remainder sets

The first contraction operator implemented is partial meet contraction. For a set A and a sentence α, the program computes the set of remainders, which are the maximal subsets of A that do not imply α.The procedure first generates every subset of A using the combinations function from itertools. Then removes the ones that are not maximal with respect to the failure to imply α.The algorithm works in three steps:

1. generate all subsets of A;
2. retain only the subsets B such that $B \nvdash \alpha$;
3. remove any subset from the collection of sets obtained in step (2.) that has a strict superset in the collection.

3.4 Partial meet contraction

Once the remainders are known, the implemented algorithm executes the following steps to compute the partial meet contraction:

1. Generate every non-empty combination of remainders in $A \perp \alpha$, ranging from combinations of a single remainder to the combination containing all remainders.
2. Intersect the remainders in each combination to obtain a candidate outcome.
3. Enumerate every such intersection and present it to the user as a candidate outcome of partial meet contraction.
4. As a special case, intersect all available remainders to obtain the full meet contraction.

If the set of remainders $A \perp \alpha$ is empty, the program produces A as the output of both the partial meet and full meet contraction operators.

3.5 Bounded contraction operator

A bounded contraction $\div$ is a contraction operator whose outcome is obtained according to the following two conditions: (i) if $A \perp \alpha = \varnothing$, then $A \div \alpha = A$; (ii) otherwise, $A \div \alpha$ is a subset of some remainder of A by α that contains a given set B (the lower

bound). To compute the bounded contraction, the implemented algorithm executes the following steps:

1. Generate the power set of each remainder of A by α.
2. Compute the union of the power sets obtained in (1).
3. Derive the candidate contraction outcomes from these subsets and keep only those that contain the chosen lower bound B.

3.6 α - Kernels

A kernel of A with respect to α is a minimal subset B that imply α (the α-kernels). The program identifies these kernels by examining every subset of A and keeping only those that still imply α while none of its strict subset does.

Hence, the algorithm works in two steps:

1. generate every subset of A;
2. retain only the subsets that still imply α while none of their subsets does.

3.7 Kernel contraction

For determining the possible outcomes of a kernel contraction by α the algorithm implements the result presented in Observation 3. Therefore, it reuses the algorithm described in Section 3.6 (Bounded contraction operator): when $A \perp \alpha \neq \varnothing$, the candidate outcomes are the sets that contain $\bigcap(A \perp \alpha)$. In this sense, kernel contraction is a bounded contraction with the lower bound $\bigcap(A \perp \alpha)$. This procedure uses the result presented in Observation 3.

3.8 Smooth kernel contraction

To compute the smooth kernel contraction, the implemented algorithm executes the following steps (when $A \perp \alpha \neq \varnothing$; otherwise, the output is A):

1. Given a belief base A and a input formula α, computes the possible outcomes of a kernel contraction by α
2. For each candidate B, inspect every element e of the original set A:
   (a) If $e \notin B$ and $B \vdash e$, then the candidate B is rejected.
   (b) If no element of $A \setminus B$ is derivable from B, then B is kept.
3. The final outcome is the collection of candidates that satisfy the filtering condition describe in (2).

In this way, the smooth kernel contraction acts as a conservative refinement of the kernel contraction: it keeps only those candidates that preserve the inferential content of the original belief base.

3.9 Basic AGM-generated base contraction

In the case of the basic AGM-generated base contraction, the computational process is structured differently, proceeding as follows:

1. Given a belief set A and a input formula α, computes the powerset of the maximal subsets of A that do not imply α.
2. Computes the union of these powerset obtained in (1)
3. Treat each such part as a candidate contraction result.
4. For each candidate subset B, inspect every element e of A that is not in B:
   (a) If $B \vdash \alpha \vee e$, then the candidate B is discarded.
   (b) If no such element e exists, then B is kept.
5. The final outcome is the list of candidates that pass through the filtering procedure described in (4).

Appendix

Executable (.exe):
https://drive.google.com/drive/folders/155ZMPmhsRcX1nufI6l11-xjsH69mZmhx?usp=drive_link

Source code:
https://github.com/FGouveia7/Belief-Bases-Program

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